

Student Career Recommendation Tool

AI-Powered Career Guidance Using Machine Learning

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INSTITUTION:- CHANDIGARH UNIVERSITY

PROGRAMME:- MCA (AI & ML)

Problem Statement & Objectives

Problem

- Students face confusion about career choices after graduation
- Lack of personalized, data-driven guidance systems
- Traditional counseling is limited in scale and objectivity
- Mismatch between skills and chosen career paths

Objectives

- Build an ML model to recommend suitable career paths
- Perform EDA to understand student data patterns
- Compare multiple ML algorithms for best accuracy
- Deliver an interactive, user-friendly recommendation system

Dataset Overview

 5550

Total Records

 21

Features

 18

Career Labels

 0 %

Missing Values

Key Features / Columns

- Academic scores (GPA, subject marks)
- Skills & extracurricular activities
- Interests & hobbies
- Personality traits / aptitude scores
- Target: Career Category / Job Role

```
Dataset Shape:  
(5550, 23)
```

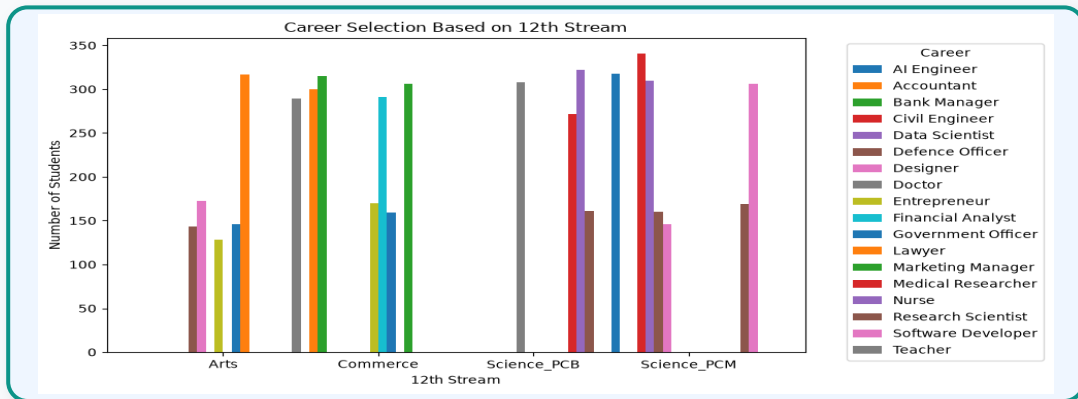
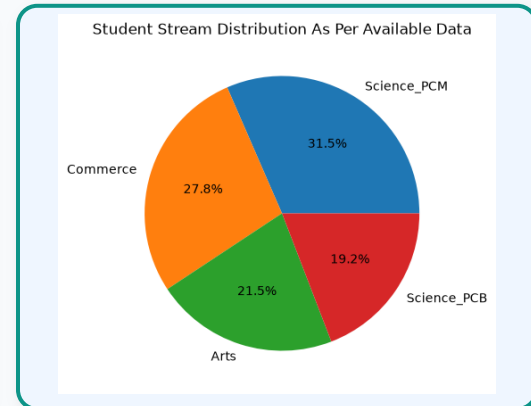
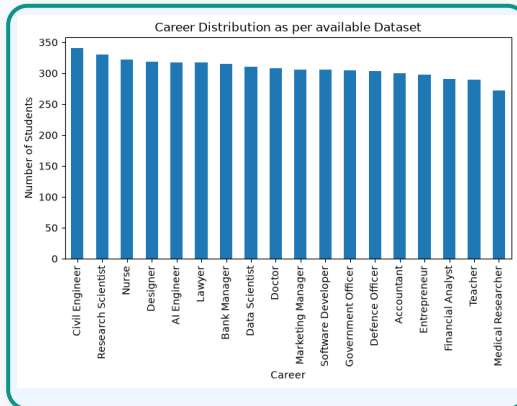
```
Columns:
```

```
Index(['Student_ID', 'Age', '10th_Percentage', '12th_Percentage',  
      '12th_Stream', 'Degree', 'CGPA', 'Programming', 'Communication',  
      'Leadership', 'Creativity', 'Problem_Solving', 'Technology_Interest',  
      'Medical_Interest', 'Finance_Interest', 'Government_Interest',  
      'Business_Interest', 'Teaching_Interest', 'Law_Interest',  
      'Physical_Fitness', 'Personality', 'Hobby', 'Career'],  
      dtype='str')
```

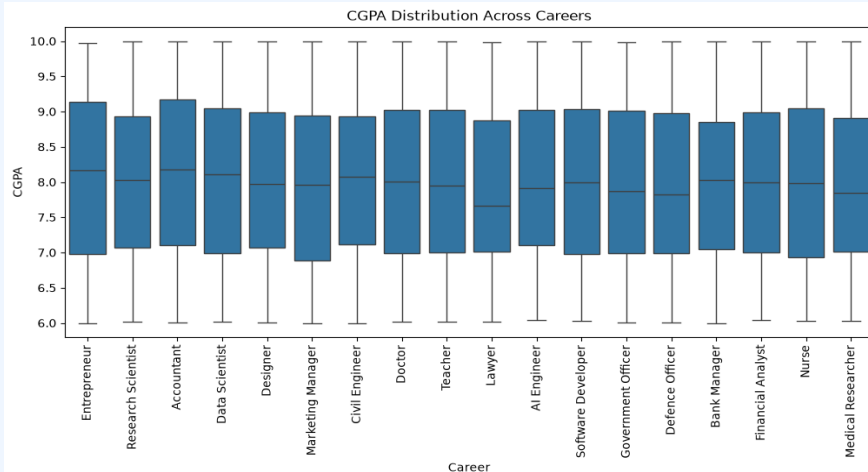
Exploratory Data Analysis — Distributions

Key Insights

- Career classes are approximately balanced, ensuring fair training across different career categories.
- Student academic features such as 10th marks, 12th marks, and CGPA show a varied distribution across the dataset.
- Career selection shows a relationship with student interests, skills, and academic background.



Exploratory Data Analysis — Feature Analysis



Data Encoding

Categorical features such as stream, degree, hobby, and personality were converted into numerical format using Label Encoding for machine learning compatibility.

Data Quality Check

- Dataset was analyzed for missing values and inconsistencies.
- No missing values were found, ensuring clean data for model training.

Feature Analysis

Student attributes such as academic scores, skills, interests, and personality were analyzed to understand their impact on career prediction.

Career Pattern Analysis

Different student profiles were compared with career categories to identify patterns between skills, interests, and recommended career paths.

Data Preprocessing Pipeline

01

Data Loading

Load the student career dataset and inspect structure, columns, and data quality.

02

Data Validation

Check missing values, data types, and ensure the dataset is ready for processing.

03

Feature Encoding

Convert categorical features in numerical columns.

04

Train-Test Split

Divide data into training and testing sets for model training and evaluation.

05

Model Preparation

Prepare processed features and target labels for machine learning algorithms

```
from sklearn.model_selection import train_test_split

X_train,X_test,y_train,y_test=train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

print("Dataset Split Summary")
print("-" * 30)
print(f"Training Features (X_train): {X_train.shape}")
print(f"Testing Features (X_test) : {X_test.shape}")
print(f"Training Labels (y_train): {y_train.shape}")
print(f"Testing Labels (y_test) : {y_test.shape}")
```

✓ 0.0s

Dataset Split Summary

Training Features (X_train): (4440, 22)

Testing Features (X_test) : (1110, 22)

Training Labels (y_train): (4440,)

Testing Labels (y_test) : (1110,)

Model Training & Algorithm Comparison

Decision Trees

Accuracy: 73.96%

Simple tree-based classifier used as baseline model.

Random Forest ?

Accuracy: 78.64%

Best performing model selected for final deployment.

KNN

Accuracy: 6%

Distance-based approach with lower performance on this dataset.

Extra Trees

Accuracy: 77.83%

Ensemble model with strong performance.

Gradient Boosting

Accuracy: 77.56%

Boosting technique with competitive results.

	Model	Accuracy
1	Random Forest	78.648649
2	Extra Trees	77.837838
3	Gradient Boosting	77.567568
0	Decision Tree	73.963964

Model Evaluation & Results

78.64%

Accuracy

80%

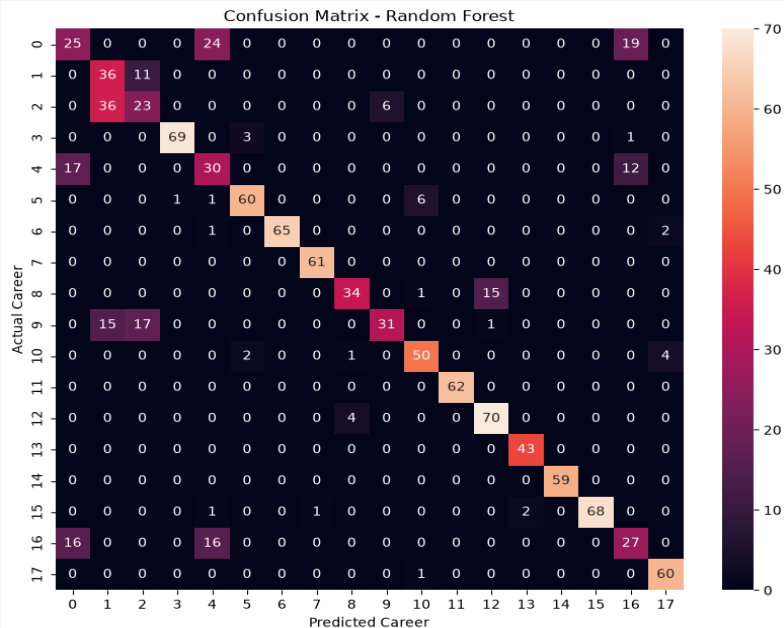
Precision

79%

Recall

79%

F1-Score

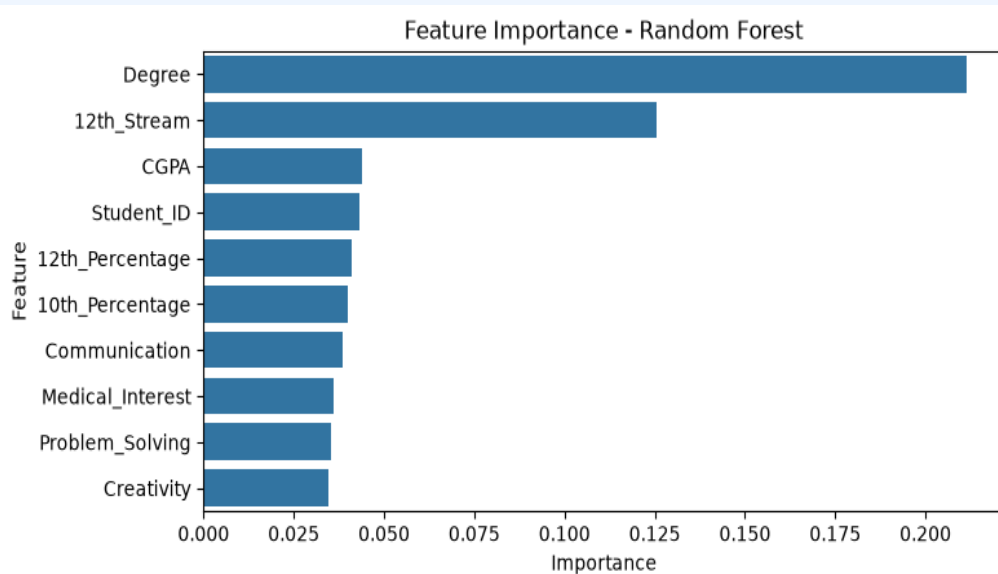


Detailed classification report

	precision	recall	f1-score	support
0	0.43	0.37	0.40	68
1	0.41	0.77	0.54	47
2	0.45	0.35	0.40	65
3	0.99	0.95	0.97	73
4	0.41	0.51	0.45	59
5	0.92	0.88	0.90	68
6	1.00	0.96	0.98	68
7	0.98	1.00	0.99	61
8	0.87	0.68	0.76	50
9	0.84	0.48	0.61	64
10	0.86	0.88	0.87	57
11	1.00	1.00	1.00	62
12	0.81	0.95	0.88	74
13	0.96	1.00	0.98	43
14	1.00	1.00	1.00	59
15	1.00	0.94	0.97	72
16	0.46	0.46	0.46	59
17	0.91	0.98	0.94	61
accuracy			0.79	1110
macro avg	0.79	0.79	0.78	1110
weighted avg	0.80	0.79	0.79	1110

Best Model & Feature Importance

🏆 Best Model: Random Forest | Test Accuracy: 79%



Top Predictive Features

- Degree: 0.2113
- 12th_Stream: 0.1255
- CGPA: 0.0438
- 12th_Percentage: 0.0409
- 10th_Percentage: 0.0400

System Demo & Sample Output

Input (Sample Student)

- Age: 21
- 12th Stream: Science_PCM
- Degree: BTech
- CGPA: 8.5
- Programming: 5/5
- Technology Interest: 5/5
- Communication: 4/5
- Personality: Analytical



Student Career Recommendation

Answer a few questions and get a personalised career suggestion.



Recommended career

Data Scientist

BTech Science_PCM Analytical Coding

 Start over

  View input data sent to model

	Student_ID	Age	10th_Percentage	12th_Percentage	12th_Stream	Degree	CGPA	Progra
0	0	21	88	85	3	16	8.5	

The deployed application accepts student profiles and generates personalized career recommendations using the trained Random Forest model.

Challenges & Future Enhancements

🔍 Project Challenges

- Building a reliable recommendation system from limited labeled student-career data.
- Selecting meaningful features from academic performance, skills, interests, and personality traits.
- Managing class imbalance across different career categories.
- Balancing model accuracy, interpretability, and deployment efficiency.
- Ensuring recommendations remain relevant across diverse student profiles.

🚀 Future Enhancements

- Improve recommendation accuracy using a larger and more diverse dataset.
- Provide the top 3 career recommendations with confidence scores.
- Enhance the **STREAMLIT** interface for a better user experience.
- Add career guidance resources and skill development suggestions.
- Collect user feedback to continuously improve model performance.

Conclusion

- ❑ Built a robust ML pipeline for student career recommendation
- ❑ Achieved 79% accuracy with Random Forest
- ❑ Identified key factors influencing career recommendations through EDA and feature importance analysis.
- ❑ Successfully deployed the application using Streamlit for real-time career recommendations.

Project Links

- 📁 GitHub Repository: <https://github.com/nksharma417/cu-task>
- 🌐 Live Streamlit Application: <https://cu-task-assignment-417.streamlit.app/>

Thank You!

Questions & Discussion Welcome 🎓